

Rishiit Sharma

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EDUCATION

Thomas Edison State University (CGPA 4.0/4.0)

September 2025

Bachelor of Arts, Mathematics

- **Relevant Coursework:** Mathematical Modeling, Multivariable Calculus, Discrete Mathematics, Microcontrollers, AC/DC Circuits
- **Early Graduation:** On track to graduate at the age of 18, due to early enrollment and credit-by-exam with AP and CLEP examinations.
- **Standardized Testing:** GRE General (170/170Q, 161/170V, taken at the age of 15).

TECHNICAL SKILLS

- **Nanofabrication Instrumentation:** RF Sputtering (PVD), Thermal Evaporation (PVD), Chemical Vapor Deposition, Photolithography, Molecular Beam Epitaxy, HV & UHV systems
- **Analog & Digital Design:** Yosys, KiCAD, Magic, Klayout, IcarusVerilog, Verilator, SPICE
- **Computational Techniques:** Finite Difference/Element Methods, Density Functional Theory, Monte Carlo Methods, Finite Difference Time Domain
- **Programming Languages:** Python, C/C++, Fortran, Julia, Verilog/SystemVerilog, VHDL, Git, Bash

RESEARCH EXPERIENCE

Miranda House, University of Delhi

July 2025 – Sept 2025

Undergraduate Researcher

- In the Multidisciplinary Research Center, assisted in the **magnetron sputtering** of tin selenide (SnSe), titanium/platinum alloy (Ti/Pt), potassium sodium niobate (KNN), with nanometer-level thickness control, to progress the state of the art in **photovoltaic cell**, **electro-optical modulation**, and **MEMS** research.
- In a three-person team, used **UV photolithography** to fabricate and wet etch **interdigital transducers (IDTs)**, and confirmed its successful fabrication under an optical microscope.
- Utilized **DFT** to identify **band structures** and **Raman/IR** frequencies for **mono, bi- and trilayer and bulk SnSe**, establishing a new state-of-the-art method for getting HSE-level accuracy with PBE, by using **+0.27eV** upward shifts (**<2% RMS error**).
- Created **Vritra**, an application in use by the center, for automated graph-generation and analysis of the **complex-valued** outputs of **impedance analyzers**, streamlining their **nanocharacterization** workflow considerably.

Defense Research and Development Organization

June 2025 – July 2025

Instrumentation Contractor

- Assisted in a vacuum pumpdown in the **III-V molecular beam epitaxy lab** at the **Solid State Physics Laboratory** inside a **Class 10,000** cleanroom, with peak vacuums of **10^{-11} Torr**, and successful homoepitaxial growth of **GaAs** at 1500 nm/hour.
- Created **Gandiva**, a bespoke **real-time RHEED analysis** tool for the **II-VI & III-V MBE labs**, developing a computer vision analysis tool **used in production** in two nanofabrication machines worth a combined **\$600,000+**.
- In parallel, working on re-enabling the **Sopra GE5SE spectroscopic ellipsometer**, by reverse engineering its embedded systems and using **PvElli** for the computational analysis, to singlehandedly re-enable a defunct piece of hardware worth **\$10,000+**.

Deshbandhu College, University of Delhi

May 2025 – June 2025

Research Assistant

- Independently synthesized multiple batches of [MBIHC](#) (an aromatic Schiff base) by reflux in acidic MeOH, and isolated via vacuum distillation with **>70% yield**.
- Confirmed its presence via thin layer chromatography, and improved the solvent mixture used from **15% EtOAc to 5%**, massively improving its visibility under short and long-wavelength UV.
- Performed [DFT studies](#) of MBIHC in **ORCA (wB97X-D3)**, improving over the published energy calculations by **29%**, and the dipole calculation by **over 740%** (calculated values are -1312.89 Hartree, 1.44 Debye).

OPEN SOURCE

Hume S1: A \$2000 Scanning Electron Microscope

<https://github.com/rolypolytoy/S1>

- An **in-progress project** based on reconstructing the SEM design in this [seminal paper](#), and so aims to provide an open-source, mass-producible, **nanometer-resolution SEM** for **\$2000**, a **>100x** improvement in cost over commercial SEMs. Expected completion is in **Q2 2025**, with all plastic components fabricated, all non-material components have arrived, and all material components are en route.
- Created a bleeding-edge [charge sensitive amplifier](#) PCB in KiCAD, using an ultra low-noise operation amplifier in a charge integrator setup, surpassing the incumbent analog processing corporation's best product (Cremat CR-113) by a **>2x reduction in cost**, **>5x bandwidth**, and **>750x voltage gain**.
- Designed a [high-vacuum capable diffusion pump](#) for **+\$10,000** savings over **turbomolecular pumps**, a **custom analog signals processing pipeline** (Phosphor P47 → SiPM → CSA → Pulse Shaper → 16-bit ADC → Microprocessor) for **\$5000+** savings in costs, and a **self-designed vacuum chamber** for **\$1000+** savings in costs as **one of dozens** of decisions made to save cost without compromising on quality.

Picht: Open-Source Electron Optics

<https://doi.org/10.5281/zenodo.15376240>

- An electron optics simulator optimized for electron microscope, mass spectrometer, and focused ion beam design and simulation, currently under **peer review** by the **Journal of Open Source Science (JOSS)** and **PyOpenSci**, with a preprint already on [Zenodo](#).
- Uses the **second-order finite difference method** for Laplace's equation, to solve for electric and magnetic field values, and **numerically integrates elliptic PDEs** to solve for the electrodynamics of **1000+ charged particles** in minutes, locally, and on any OS.
- Has a comprehensive [documentation website](#), CI/CD on [Github Actions](#), 200+ commits on [GitHub](#), and 25+ releases on [PyPi](#).

Synthesizing the IBM A2O CPU

<https://github.com/OpenPOWERFoundation/a2o>

- Found and fixed a critical synthesis issue related to **clock width logic**, and demonstrated its efficacy by synthesizing a Verilog netlist with **>500,000** logic gates using the **SKY130 PDK**.
 - Made a [pull request](#) that was merged into the master branch of the **A2O** repository, becoming the **fourth contributor** to the **most powerful open-source Verilog CPU**.
 - Published the completed netlists to a public-facing [repository](#), permanently reducing the barrier of entry to the **open-source VLSI** community.
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